



Year 11

Mathematics Extension 1

Assessment Task 3

2018

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided with this paper
- In Questions 11 – 13, show relevant mathematical reasoning and/or calculations

Total marks – 55

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.
- Answer on the provided Multiple Choice Answer Sheet.

Section II

45 marks

- Attempt Questions 11 – 13
- Allow about 75 minutes for this section

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which one of the following functions is *not* even?

A) $f(x) = x^2$

B) $f(x) = \frac{1}{1+x^2}$

C) $f(x) = \sqrt{4-x^2}$ when $|x| \leq 2$

D) $f(x) = x\sqrt{1-x^2}$ when $|x| \leq 1$

2 State the natural domain of $y = \frac{1}{\sqrt{4-x^2}}$:

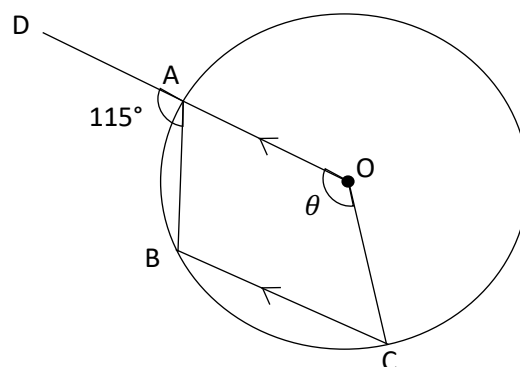
A) Domain: $\{x : -2 < x < 2\}$

B) Domain: $\{x : -2 \leq x \leq 2\}$

C) Domain: $\{x : x < -2, x > 2\}$

D) Domain: $\{x : x < \pm 2\}$

- 3 In how many ways can 5 girls and 3 boys be arranged in a row if there are 7 girls and 4 boys to select from?
- A) 84
- B) 60 480
- C) 3 386 880
- D) 2 438 553 600
- 4 How many points on the curve of the function $y = \frac{1}{x-1} - \frac{4}{x-4}$ have a gradient of zero?
- A) 0
- B) 1
- C) 2
- D) 4
- 5 The diagram shows a circle with centre O.
- A, B and C are points on the circle, and OA is produced to D.
- DO is parallel to BC, and $\angle DAB = 115^\circ$.



Find the size of angle θ .

- A) 100°

B) 115°

C) 130°

D) 230°

6 If $t = \tan \frac{1}{2}\theta$ which of the following expressions is equivalent to $\sec \theta$?

A) $\frac{1+t^2}{1-t^2}$

B) $\frac{2t}{1+t^2}$

C) $\frac{1-t^2}{1+t^2}$

D) $\frac{2t}{1-t^2}$

7 If θ is an acute angle, where $\sin \theta = \frac{1}{\sqrt{5}}$, find the exact value of $\sin 2\theta$.

A) $\frac{4}{5}$

B) $\frac{2}{\sqrt{5}}$

C) $\frac{1}{2\sqrt{5}}$

D) $\frac{2}{5}$

8 Which of the following is an expression for $\cos(A-B) - \cos(A+B)$

A) $-2\cos A \cos B$

B) $-2 \sin A \sin B$

C) $2 \cos A \cos B$

D) $2 \sin A \sin B$

9 Which of the following is *not* a function?

A) $y = \sqrt{x}$

B) $x = \sqrt{y}$

C) $y = 2$

D) $x = 2$

10 $2^x \times 5^x$ is equal to:

A) 7^x

B) 7^{2x}

C) 10^x

D) 10^{2x}

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Section II

45 marks

Attempt Questions 11–13

Allow about 75 minutes for this section

Answer each question in a **separate writing booklet**. Extra writing booklets are available.

In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 11: 15 marks

- a) Find the exact value $\cos 105^\circ$ 2
- b) Differentiate $y = \frac{\sqrt{x+1}}{x}$, leaving your answer in simplest fraction form – ie. 3
As a single fraction.
- c) Solve $\frac{2x+5}{x+3} \leq 1$. 2
- d) i) Show that $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ when $\cos 2\theta \neq -1$. 3
- ii. Hence show that the exact value of $\tan 15^\circ$ is $2 - \sqrt{3}$. 2
- e) i. In how many ways can 6 friends sit around a circular table? 1
- ii. In how many ways can the 6 friends sit around a circular table if 2 of the friends insist on sitting next to each other? 1

- iii. In how many ways can the 6 friends sit around a circular table if 2 of the friends have had a fight and refuse to sit next to each other. **1**

End of question 11

Question 12: 15 marks – Start a new booklet

- a) Consider the function $y = \frac{x}{2-x}$
- State the equation of the horizontal asymptote. **1**
 - Sketch the graph of the function including all intercepts and asymptotes **2**
- b) How many distinct arrangements can be made by using the letters of the word POSSIBILITY:
- With no restrictions? **1**
 - If the letter **s** does not occupy the first or last place? **2**
 - If the letters **s** are together? **1**
- c) Vanessa needs to choose five representatives from the nine exchange students at her school. **2**
- The exchange students consist of 4 Americans, 2 Japanese and 3 Germans.
- In how many ways can she select the representatives if she wishes to include at least one student of each nationality?
- d) The point $P(x, y)$ divides the line joining the points $A(5, -2)$ and $B(-1, -5)$ externally in the ratio 1:2. Find the coordinates of P . **2**

e) i. Sketch the graph of $y = |2x + 1|$. **1**

ii. Hence, or otherwise, solve $|2x + 1| \geq |x - 2|$. **3**

End of question 12

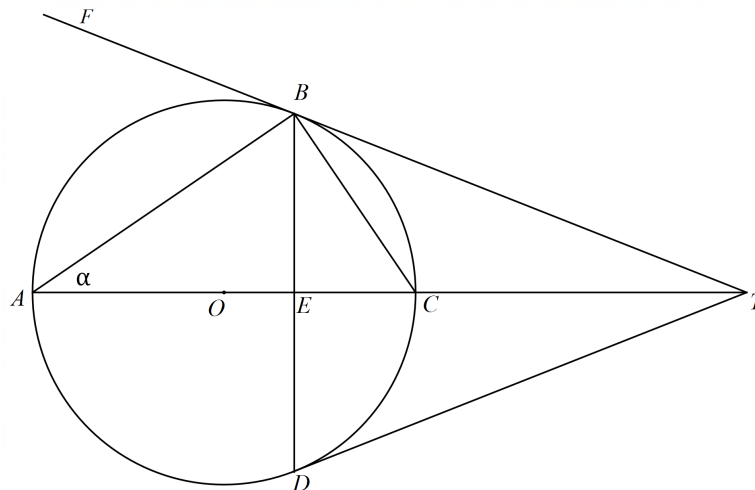
Question 13: 15 marks – Start a new booklet

- a) The point O is the centre of the circle $ABCD$. FT is a tangent at B . DT is a tangent at D .

The points A , O , E , C and T are collinear.

$BD \perp AT$ and $\angle BAE = \alpha$.

Copy the diagram into your booklet.

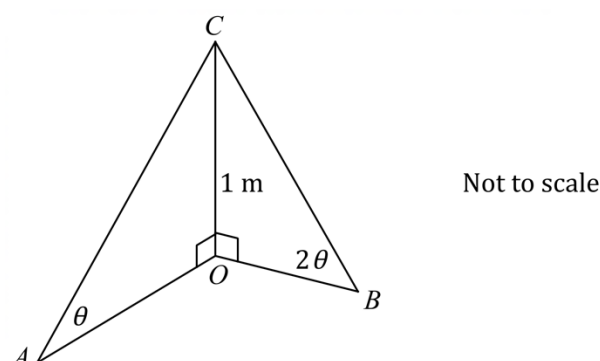


- i. Explain why $\angle CBT = \alpha$. **1**

- ii. Prove that BC bisects $\angle EBT$. 2
- iii. Prove that AB bisects $\angle EBF$. 1
- b) The curves $y = x^2$ and $y = x^2 - 3x + 6$ meet at the point $(2, 4)$. What is the acute angle between the tangents to the curves at this point, to the nearest minute? 3

Question 13 continues on the next page

- c) The diagram below shows a vertical pole OC of height 1 metre. It stands on horizontal ground. The angles of elevation of the top (C) of the pole, from points A and B , on the ground are θ and 2θ respectively.



- i. Show that $OA = \cot \theta$, $OB = \cot 2\theta$ and $BC = \operatorname{cosec} 2\theta$ 1
- ii. Hence or otherwise, show that $OA - OB = BC$ 2
- d) i. Express $\sin(2x) - \cos(2x)$ in the form $R \sin(\theta - \alpha)$ 2
- ii. Hence or otherwise, solve the equation $\sin(2x) - \cos(2x) = 1$ on the domain $0 \leq x \leq 360^\circ$. 3

End of examination



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Section II

45 marks

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Section I

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Use the multiple-choice answer sheet for Questions 1–10.

1 Which one of the following functions is *not* even?

A) $f(x) = x^2$

B) $f(x) = \frac{1}{1+x^2}$

C) $f(x) = \sqrt{4-x^2}$ when $|x| \leq 2$

☒ D) $f(x) = x\sqrt{1-x^2}$ when $|x| \leq 1$ $f(x) = x\sqrt{1-x^2}$ when $|x| \leq 1$

$$f(-x) = -x\sqrt{1-(-x^2)}$$

$$f(-x) = -x\sqrt{1-x^2} = -(x\sqrt{1-x^2})$$

$$= -f(x)$$

$\therefore$ it is an odd function

D

2 State the natural domain and range of $y = \frac{1}{\sqrt{4-x^2}}$:

☒ A) Domain: $\{x: -2 < x < 2\}$

B) Domain: $\{x: -2 \leq x \leq 2\}$

C) Domain: $\{x: x < -2, x > 2\}$

D) Domain: $\{x: x < \pm 2\}$

A

- 3 In how many ways can 5 girls and 3 boys be arranged in a row if there are 7 girls and 4 boys to select from?
- A) 84
- B) 60 480
- ☒ C) 3 386 880
- D) 2 438 553 600

5 girls can be selected from 7 girls in 7C_5 ways.

C

3 boys can be selected from 4 boys in 4C_3 ways.

Each selected group can then be arranged in $8!$ ways.

Total arrangements = ${}^7C_5 \cdot {}^4C_3 \cdot 8!$

$$= 3\,386\,880$$

- 4 How many points on the curve $y = \frac{1}{x-1} - \frac{4}{x-4}$ have a gradient of zero?

- A) 0
- B) 1
- ☒ C) 2
- D) 4

C

$$y = \frac{1}{x-1} - \frac{4}{x-4}$$

C

$$y = (x-1)^{-1} - 4(x-4)^{-1}$$

$$y' = \frac{-1}{(x-1)^2} + \frac{4}{(x-4)^2}$$

$$y' = \frac{-(x-4)^2 + 4(x-1)^2}{(x-1)^2(x-4)^2}$$

$$y' = \frac{-x^2 + 8x - 16 + 4x^2 - 8x}{(x-1)^2(x-4)^2}$$

$$y' = \frac{3x^2 - 12}{(x-1)^2(x-4)^2}$$

$$3x^2 - 12 = 0 \text{ for stationary point}$$

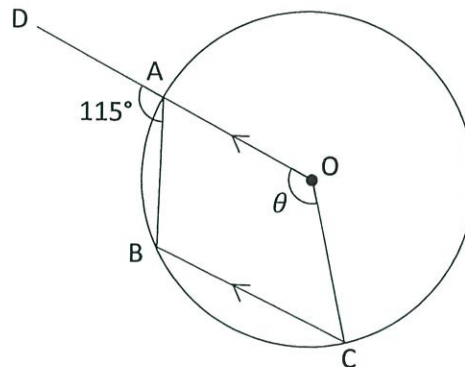
$$3(x^2 - 4) = 0$$

$$x = \pm 2 (\therefore 2 \text{ stat points})$$

- 5 The diagram shows a circle with centre O.

A, B and C are points on the circle, and OA is produced to D.

DO is parallel to BC, and $\angle DAB = 115^\circ$.



Find the size of angle θ .

A) 100°

B) 115°

☒ C) 130°

D) 230°

$\angle ABC = 115^\circ$ (alternate angles
on parallel lines)

$\text{reflex} \angle AOC = 230^\circ$ (reflex angle
at the centre is 2 times the angle
subtending the same major arc)

$$\theta = 360^\circ - 230^\circ = 130^\circ$$

C

6 If $t = \tan \frac{1}{2} \theta$ which of the following expressions is equivalent to $\sec \theta$?

- (A) $\frac{1+t^2}{1-t^2}$
B) $\frac{2t}{1+t^2}$
C) $\frac{1-t^2}{1+t^2}$
D) $\frac{2t}{1-t^2}$

A

7 If θ is an acute angle, where $\sin \theta = \frac{1}{\sqrt{5}}$, find the exact value of $\sin 2\theta$.

- (A) $\frac{4}{5}$
B) $\frac{2}{\sqrt{5}}$
C) $\frac{1}{2\sqrt{5}}$
D) $\frac{2}{5}$

$$\sin \theta = \frac{1}{\sqrt{5}} \rightarrow$$

Pythagorean theorem: $b^2 = \sqrt{5}^2 - 1^2 = 4 \rightarrow b = 2$

Double angle identity:

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \cdot \frac{1}{\sqrt{5}} \cdot \frac{2}{\sqrt{5}} = \frac{4}{5}$$

A

- 8 Which of the following is an expression for $\cos(A-B) - \cos(A+B)$

$$\cos(A-B) - \cos(A+B)$$

$$= \cos A \cos B + \sin A \sin B - (\cos A \cos B - \sin A \sin B)$$

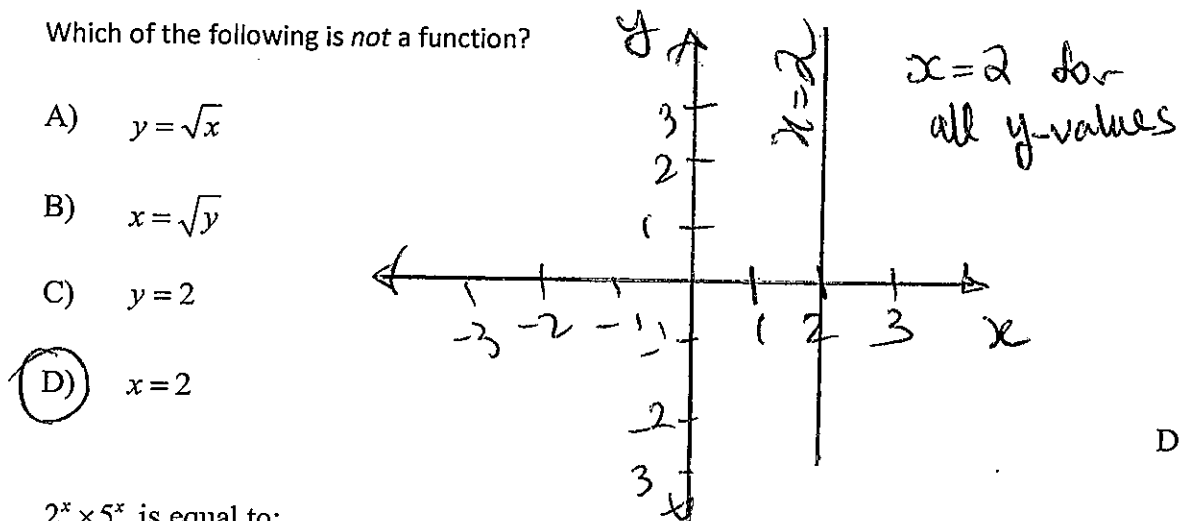
$$= \cos A \cos B + \sin A \sin B - \cos A \cos B + \sin A \sin B$$

$$= 2 \sin A \sin B$$

A) $-2 \cos A \cos B$
 B) $-2 \sin A \sin B$
 C) $2 \cos A \cos B$
 D) $2 \sin A \sin B$

D

- 9 Which of the following is *not* a function?



D

- 10 $2^x \times 5^x$ is equal to:

A) 7^x
 B) 7^{2x}
 C) 10^x
 D) 10^{2x}

$$2^x \times 5^x = (2 \times 5)^x$$

$$= 10^x$$

C

Section II

45 marks

Attempt Questions 11–13

Allow about 75 minutes for this section

Answer each question in a **separate writing booklet**. Extra writing booklets are available.

In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 11

- a) Find the exact value $\cos 105^\circ$

2

$$\cos 105^\circ = \cos (60^\circ + 45^\circ)$$

$$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$$\begin{aligned} &\text{or } \frac{1 - \sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4} \end{aligned}$$

- b) Differentiate $y = \frac{\sqrt{x+1}}{x}$, leaving your answer in simplest fraction form – ie. As a single fraction.

3

$$u = (x+1)^{\frac{1}{2}}, v = x$$

$$u' = \frac{1}{2}(x+1)^{-\frac{1}{2}}, v' = 1$$

$$u' = \frac{1}{2\sqrt{x+1}}$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$y' = \frac{x \cdot \frac{1}{2\sqrt{x+1}} - \sqrt{x+1}}{x^2}$$

$$y' = \frac{\frac{x}{2\sqrt{x+1}} - \sqrt{x+1}}{x^2}$$

$$y' = \frac{x - \sqrt{x+1} \cdot 2\sqrt{x+1}}{2\sqrt{x+1} x^2}$$

$$y' = \frac{x - 2(x+1)}{2\sqrt{x+1}} \div x^2$$

$$y' = \frac{x - 2x - 2}{2\sqrt{x+1}} \times \frac{1}{x^2}$$

$$y' = \frac{-x - 2}{2x^2\sqrt{x+1}}$$

$$\text{or } y' = -\frac{x+2}{2x^2\sqrt{x+1}}$$

c) Solve $\frac{2x+5}{x+3} \leq 1$.

or $x \neq -3$

2

$$\frac{2x+5}{x+3} \leq 1$$

$$2x+5 = x+3$$

$$x = -2$$

$$(x+3)^2 \frac{2x+5}{x+3} \leq 1(x+3)^2$$

Test:

$$x = -4$$

$$x = 0$$

$$x = -2.5$$

$$(x+3)(2x+5) \leq (x+3)^2$$

$$2x^2 + 11x + 15 \leq x^2 + 6x + 9$$

$$\frac{2x-4+5}{-4+3} \leq 1$$

$$\frac{0+5}{0+3} \leq 1$$

$$\frac{2x(-2.5)+5}{-2.5+3} \leq 1$$

$$x^2 + 5x + 6 \leq 0$$

$$\frac{-3}{-1} \leq 1$$

false

$$0 \leq 1$$

$$(x+3)(x+2) \leq 0 \text{ test a point}$$

$$-3 < x \leq -2$$

False

$$\therefore -3 \leq x \leq -2$$

True

d) i)

Show that $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ when $\cos 2\theta \neq -1$.

3

$$\begin{aligned} LHS &= \frac{1 - (\cos^2 \theta - \sin^2 \theta)}{1 + (\cos^2 \theta - \sin^2 \theta)} \\ &= \frac{1 - \cos^2 \theta + \sin^2 \theta}{1 + \cos^2 \theta - \sin^2 \theta} \text{ divide through by } \cos^2 \theta \\ &= \frac{\sec^2 \theta - 1 + \tan^2 \theta}{\sec^2 \theta + 1 - \tan^2 \theta} \\ &= \frac{(\tan^2 \theta + 1) - 1 + \tan^2 \theta}{(\tan^2 \theta + 1) + 1 - \tan^2 \theta} \\ &= \frac{2\tan^2 \theta}{2} \\ &= \tan^2 \theta \\ &= RHS \end{aligned}$$

3 marks for proof reaching the required result.

2 marks for a proof with one or two minor errors or which doesn't quite reach the required result

1 mark for a proof with some correct working towards result

ii.

Hence show that the exact value of $\tan 15^\circ$ is $2 - \sqrt{3}$.

2

$$\tan^2(15^\circ) = \frac{1 - \cos 2(15^\circ)}{1 + \cos 2(15^\circ)}$$

$$\tan^2(15^\circ) = \frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}}$$

$$\tan^2(15^\circ) = \frac{2 - \sqrt{3}}{2 + \sqrt{3}}$$

$$\tan^2(15^\circ) = \frac{2 - \sqrt{3}}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}$$

$$\tan^2(15^\circ) = \frac{(2 - \sqrt{3})^2}{2^2 - (\sqrt{3})^2}$$

$$\tan^2(15^\circ) = \frac{(2 - \sqrt{3})^2}{4 - 3}$$

$$\tan^2(15^\circ) = (2 - \sqrt{3})^2$$

Since $\tan 15^\circ$ is positive (1st quadrant)

$$\tan(15^\circ) = 2 - \sqrt{3}$$

2 marks for the correct solution

1 mark for use of the derived formula with an error resulting in an incorrect result

- e) i. In how many ways can 6 friends sit around a circular table?

1

$$(n - 1)! = (6 - 1)! = 5! = 120$$

- ii. In how many ways can the 6 friends sit around a circular table if 2 of the friends insist on sitting next to each other?

1

There are $2!$ ways for 2 friends to sit next to each other.

There are $4!$ ways for the rest of the friends to sit next to each other.

$$2! \cdot 4! = 48$$

- iii. In how many ways can the 6 friends sit around a circular table if 2 of the friends have had a fight and refuse to sit next to each other?

1

There are 120 ways for the 6 friends to sit. 48 involve 2 sitting next to each other.

$$120 - 48 = 72$$

Question 12:

a)

Consider the function $y = \frac{x}{2-x}$

i. State the equation of the horizontal asymptote

1

$$y = \frac{x}{2-x} = \frac{x}{-(x-2)} = \frac{x-2+2}{-(x-2)} = -\frac{(x-2)+2}{(x-2)} = -\left(\frac{x-2}{x-2} + \frac{2}{x-2}\right)$$

$$= -\frac{2}{x-2} - 1, \text{ which is a hyperbola, reflected vertically,}$$

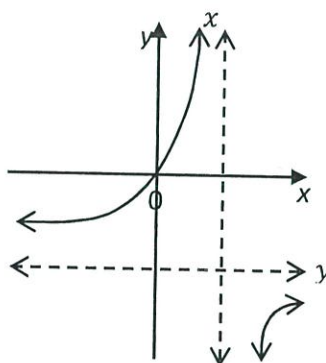
$$\text{shifted right 2 and down 1 unit.}$$

The horizontal asymptote is $y = -1$

ii. Sketch the graph of the function including all intercepts and asymptotes

2

When $x = 0, y = 0$



1 for intercepts (0,0)
1 for vertical asymptote and correct curve

b) How many distinct arrangements can be made by using the letters of the word **possibility**:

i. With no restrictions?

1

$$\text{Arrangements} = \frac{11!}{2!3!} = 3\,326\,400$$

ii. If the letter *s* does *not* occupy the first or last place?

2

Number of ways first and last place can be filled = $9 \times 8 = 72$

$$\text{Total arrangements} = \frac{72 \times 9!}{2! \times 3!} = 2\,177\,280$$

2 marks for correct answer

1 mark for a solution which includes some correct and relevant working toward an answer.

iii. If the letters s are together?

1

Treat the 2 s 's as one letter, so there are 10 letters with one letter repeated 3 times.

$$\text{Arrangements} = \frac{10!}{3!} = 604800$$

c) Vanessa needs to choose five representatives from the nine exchange students at her school. 2

The exchange students consist of 4 Americans, 2 Japanese and 3 Germans.

In how many ways can she select the representatives if she wishes to include at least one student of each nationality?

$$\begin{aligned} \text{Selections} &= {}^4C_1 \times {}^2C_2 \times {}^3C_2 + {}^4C_1 \times {}^2C_1 \times {}^3C_3 + {}^4C_2 \times {}^2C_1 \times {}^3C_2 \\ &\quad + {}^4C_2 \times {}^2C_2 \times {}^3C_1 + {}^4C_3 \times {}^2C_1 \times {}^3C_1 \\ &= 12 + 8 + 36 + 18 + 24 \\ &= 98 \end{aligned}$$

2 marks for correct solution

1 mark for working which includes some terms missing, or incorrect but has basic logic correct or if the calculation is done incorrectly

d) The point $P(x, y)$ divides the line joining the points $A(5, -2)$ and $B(-1, -5)$ externally in the ratio 1:2. Find the coordinates of P . 2

$$\begin{array}{cc} (x_1, y_1) & (x_2, y_2) \\ (5, -2) & (-1, -5) \\ m:n & \\ -1:2 & \end{array}$$

$$P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

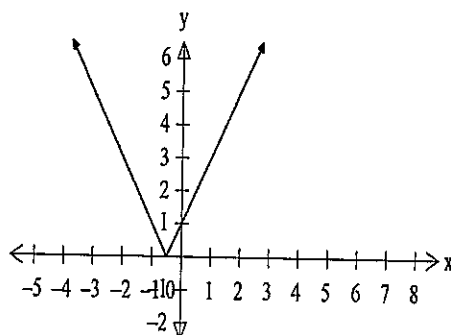
$$P\left(\frac{(-1)(-1) + 2(5)}{-1+2}, \frac{(-1)(-5) + 2(-2)}{-1+2}\right)$$

$$P\left(\frac{1+10}{1}, \frac{5-4}{1}\right)$$

$$P(11, 1)$$

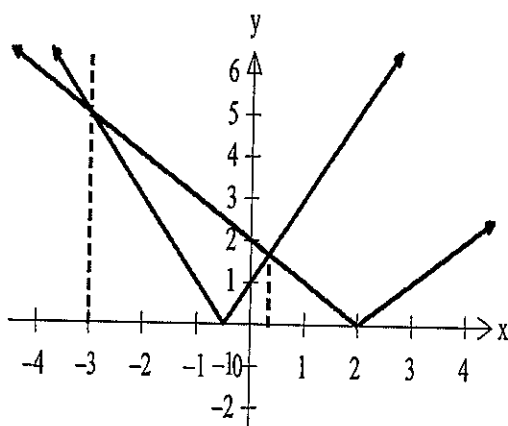
- e) i. Sketch the graph of $y = |2x + 1|$.

1



- ii. Hence, or otherwise, solve $|2x + 1| \geq |x - 2|$.

3



$$2x + 1 = x - 2 \quad \text{or} \quad 2x + 1 = -x + 2$$

$$x = -3 \qquad x = \frac{1}{3}$$

$\therefore$ from the graph and equation $x \geq \frac{1}{3}$ and $x \leq -3$

3 marks for correct solution by any method.

2 marks for a worked solution with a minor error or which not quite complete

1 mark for an attempt with some correct working towards result

Test: $x = -4$

$x = 0$

$x = 1$

$$|2(-4) + 1| \geq |-4 - 2|, \quad |2(0) + 1| \geq |0 - 2|, \quad |2(1) + 1| \geq |1 - 2|$$

$$|-8 + 1| \geq |-6|$$

$$1 \geq 2$$

$$3 \geq 1$$

$$7 \geq 6$$

False

True

True

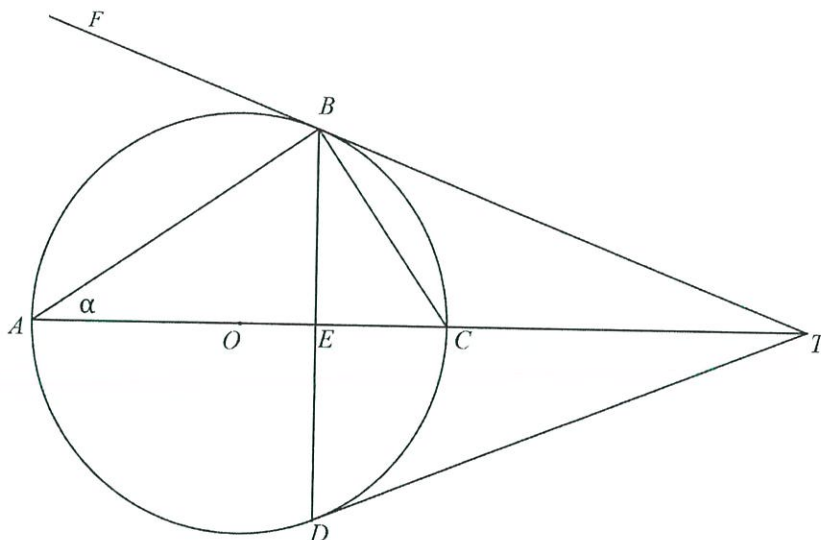
$$\therefore x \leq -3 \quad \text{or} \quad x \geq \frac{1}{3}$$

Question 13:

- a) The point O is the centre of the circle $ABCD$. FT is a tangent at B . DT is a tangent at D .

The points A , O , E , C and T are collinear.

$BD \perp AT$ and $\angle BAE = \alpha$.



- i. Explain why $\angle CBT = \alpha$.

1

$\angle CBT = \angle BAC = \alpha$ (ang in alternate seg on chord BC)

- ii. Prove that BC bisects $\angle EBT$.

2

$BD \perp AT$ ($\triangle EBT \equiv \triangle EDT$)

$\therefore \angle BET = 90^\circ$

$\angle ABC = 90^\circ$ ($\angle$ in semi circle)

$\angle ACB = 180^\circ - (90^\circ + \alpha)$ ($\angle$ sum of $\triangle ABC$)
 $= 90^\circ - \alpha$

$\therefore \angle EBC = 180^\circ - (90^\circ + 90^\circ - \alpha)$ ($\angle$ sum of $\triangle BCE$)
 $= \alpha$

since $\angle CBT = \angle EBC = \alpha$

BC bisects $\angle EBT$

2 marks for correct solution.

1 mark for a fully worked solution with a minor error in logic or calculation or for a solution which is incomplete but includes working of merit.

iii Prove that AB bisects $\angle EBF$.

1

$$\angle ABF = \angle ECB = 90^\circ - \alpha \text{ (}\angle \text{ in alt seg on chord } AB\text{)}$$

$$\angle ABE = 90^\circ - \alpha \text{ (proved above)}$$

$\therefore AB$ bisects $\angle EBF$

- b) The curves $y = x^2$ and $y = x^2 - 3x + 6$ meet at the point $(2, 4)$. What is the acute angle between the tangents to curves at this point to the nearest minute?

3

$$y = x^2$$

$$y' = 2x$$

$$\therefore m_1 = 2 \times 2$$

$$m_1 = 4$$

$$y = x^2 - 3x + 6$$

$$y' = 2x - 3$$

$$\therefore m_2 = 2 \times 2 - 3$$

$$m_2 = 1$$

1 mark for gradients

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\therefore \theta = 30.9637$$

①

$$\theta = 30^\circ 57' 49''$$

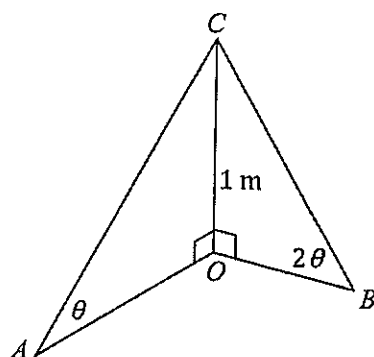
$$\tan \theta = \frac{4 - 1}{1 + 4}$$

$$\theta = 30^\circ 58'$$

①

$$\tan \theta = \frac{3}{5}$$

- c) The diagram below shows a vertical flagpole OC of height 1 metre. It stands on horizontal ground. The angles of elevation of the top (C) of the flagpole, from points A and B , on the ground are θ and 2θ respectively.



Not to scale

- i. Show that $OA = \cot \theta$, $OB = \cot 2\theta$ and $BC = \operatorname{cosec} 2\theta$

1

$$\tan \theta = \frac{1}{OA}$$

$$\therefore OA = \cot \theta$$

$$\tan 2\theta = \frac{1}{OB}$$

$$\therefore OB = \cot 2\theta$$

$$\sin 2\theta = \frac{1}{BC}$$

$$BC = \operatorname{cosec} 2\theta$$

- ii. Hence or otherwise, show that $OA - OB = BC$

2

$$\begin{aligned}
 OA - OB &= \cot\theta - \cot 2\theta \\
 &= \frac{\cos\theta}{\sin\theta} - \frac{\cos 2\theta}{\sin 2\theta} \\
 &= \frac{\cos\theta}{\sin\theta} - \frac{\cos 2\theta}{2\sin\theta\cos\theta} \\
 &= \frac{2\cos^2\theta - (\cos^2\theta - \sin^2\theta)}{2\sin\theta\cos\theta} \\
 &= \frac{\cos^2\theta + \sin^2\theta}{2\sin\theta\cos\theta} \\
 &= \frac{1}{\sin 2\theta} \\
 &= \operatorname{cosec} 2\theta \\
 &= BC
 \end{aligned}$$

2 marks: Correct answer.

1 mark: Simplifies the expression by using at least one double angle formula.

- d) i. Express $\sin(2x) - \cos(2x)$ in the form $R \sin(\theta - \alpha)$

3

$\sin(2x) - \cos(2x) = R \sin(2x - \alpha)$ (we can let the auxiliary angle's sign be determined by the

resulting triangle)

Dividing both sides by R:

$$\frac{1}{R} \sin(2x) - \frac{1}{R} \cos(2x) = \sin(2x - \alpha)$$

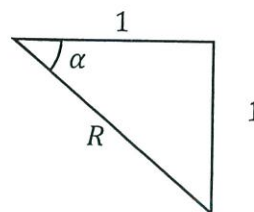
Using angle sum identity on RHS:

$$\frac{1}{R} \sin(2x) - \frac{1}{R} \cos(2x) = \sin(2x) \cos \alpha + \cos(2x) \sin \alpha$$

Equating the coefficients of $\sin(2x)$ and $\cos(2x)$:

$$\sin \alpha = \frac{1}{R}$$

$$\cos \alpha = \frac{1}{R}$$



$$R = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$$

$$\tan \alpha = \frac{1}{1} \rightarrow \alpha = 45^\circ$$

$$\sin(2x) - \cos(2x) = \sqrt{2} \sin(2x - 45^\circ)$$

- ii. Hence or otherwise, solve the equation $\sin(2x) - \cos(2x) = 1$ on the domain $0 \leq x \leq 360^\circ$.

$$\sin(2x) - \cos(2x) = 1 \rightarrow \sqrt{2} \sin(2x - 45^\circ) = 1 \text{ (by part (i))}$$

Adjusting the domain:

$$0 \leq x \leq 360^\circ \rightarrow 0 \leq 2x \leq 720^\circ \rightarrow -45^\circ \leq 2x - 45^\circ \leq 675^\circ$$

$$\sin(2x - 45^\circ) = \frac{1}{\sqrt{2}} \text{ and } \sin \theta = \frac{1}{\sqrt{2}} \text{ at } \theta = 45, 135^\circ, \dots \text{ (sine is}$$

positive in
Quadrants
I and II)

$$2x - 45^\circ = 45^\circ, 135^\circ, 405^\circ, 495^\circ$$

$$2x = 90^\circ, 180^\circ, 450^\circ, 540^\circ$$

$$x = 45^\circ, 90^\circ, 225^\circ, 270^\circ$$

